Analysis of Zero Stiffness Permanent Magnet Bearing Including Reluctance Effect

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The isolation bandwidth of zero stiffness permanent magnet bearing (ZSPMB) is limited by the stiffness variation near the zero stiffness point. This paper proposes an accurate and effective modeling technique to analyze and optimize the stiffness of ZSPMB in working range. Since the stiffness of ZSPMB is very sensitive to modeling errors, the variable current model of the permanent magnet is first derived considering magnetic permeability, the influences of which on the magnetic field and force are attributed to the reluctance effect in this paper. Then, the force exerted between permanent magnets (PMs) is equivalently decomposed into Lorentz force and reluctance force. Based on the Biot–Savart law and elliptic integral, the magnetic field of rings with axial, radial magnetizations and the equivalent Lorentz force (ELF) of typical ZSPMB topologies are calculated analytically. The equivalent reluctance force (ERF) is subsequently obtained by combining the finite element method. Compared with the ELF, the ERF is about one order of magnitude smaller and in the opposite direction and it causes the stiffness-displacement curve to drift and deflect. Finally, a Halbach ZSPMB is optimized with the ELF model, and then the ERF is suppressed by fine-tuning the structural size. The finite element analysis results show that the stiffness variation in the working range is effectively reduced, and the validity of the proposed hybrid analytical-finite element model modeling technique is verified.

Index Terms— **Halbach, hybrid analytical-finite element model (FEM) approach, permanent magnet (PM) bearing, reluctance effect, zero stiffness.**

I. Introduction

Many high-performance mechatronic systems are becoming ever more demanding in terms of vibration isolation bandwidth [1]. Reducing the bearing stiffness is beneficial for decreasing the resonance frequency of the system and increasing the isolation bandwidth [2]. Zero stiffness permanent magnet bearing (ZSPMB) can directly utilize magnetic force to achieve a substantially constant gravity compensation capability over a limited working range. Compared with the conventional low-stiffness bearing based elastic material, the ZSPMB has advantages such as no static deformation, no contact, simple structure, and so on. It becomes a feasible alternative technique for providing an ultra-quiet dynamic environment for sensitive apparatuses.

Related studies on permanent magnet (PM) bearings have often been aimed at generating great force with high stiffness [1], [3]. In [3], the axial force and stiffness of bearings with axial, radial, and perpendicular magnetizations were compared, and the bearings made of stacked ring magnets with alternate magnetizations were studied for identifying the structures required for having great axial forces and stiffness. The results also indicate that zero stiffness can be obtained at extreme points of force. Several authors investigated topologies with low stiffness [1], [2], [4]. In [2], zero stiffness is achieved by a sandwiched structure, where the middle magnet is attracted to the upper magnet and repelled by the bottom magnet. In [4], a cylindrical Halbach magnetic levitation gravity compensator was proposed for the fine stage of the high-precision positioning system. In [1], a stacked PMs vibration isolation system with 730 kg bearing capacity was developed. Because the force and stiffness of PM bearings are highly position dependent and strongly coupled with each other, the localized zero stiffness is valid only for small variations in the gaps between the magnets [2]. Hence, the isolation bandwidth of ZSPMB is limited by the stiffness variation near the zero stiffness point.

Accurate calculation of the magnetic field and force of ZSPMB is extremely important for its performances prediction and structure design. Numerical methods are able to model complex geometries and nonlinearities accurately. However, due to the low stiffness characteristic of ZSPMB, a smooth force-displacement curve is needed to avoid the stiffness calculation error caused by the derivative. Fine mesh can improve the accuracy of the numerical method, but it also makes the calculation very time-consuming. The current model [4], [5] and the charge model [1]–[3], [5]–[7] are commonly used as analytical solutions for cuboidal magnets and magnetic rings. To reduce the number of numerical integration and improve the computational efficiency, the volume charge density is ignored for the radial magnetized ring magnet, and the elliptic integral is used [3], [6]. The analytical models are more suitable for parametric studies and optimization, but the nonlinear effect of PM relative permeability has often been ignored. Since the stiffness of ZSPMB is highly sensitive to modeling errors, in [4], the zero stiffness position drift caused by PM operating point was found and an improved current model was established using the average operating point obtained by magnetostatic field simulation. The following analysis in this paper shows that the PM relative permeability results in the drift and deflection of the stiffness-displacement curve, which is the key factor affecting the ZSPMB stiffness variation in the working range. However, the model based on the

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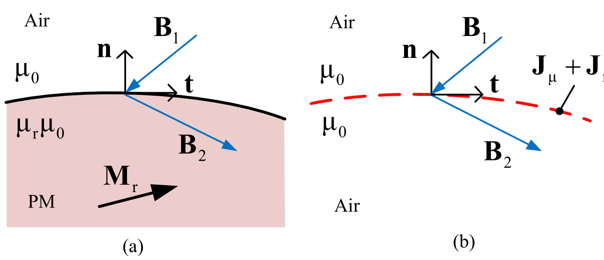

Fig. 1. Variable current model of PM considering magnetic permeability. (a) PM. (b) Equivalent current.

average operating point is still an approximate description of the nonlinear effect. In [7], a variable surface charge density model was built; however, the numerical iterative model has an undesirable long computational time. Therefore, an accurate and effective modeling technique including nonlinear effect is necessary for the analysis of ZSPMB.

Attributing the influences of magnetic permeability on the magnetic field and force to the reluctance effect, this paper proposes a hybrid analytical-finite element model (FEM) approach for the analysis and optimization of the ZSPMB. First, based on the variable current model of PM, the force exerted between PMs is equivalently decomposed into Lorentz force and reluctance force. Then, applying the Biot–Savart law and elliptic integral, the magnetic field of rings with axial, radial magnetizations and the equivalent Lorentz force (ELF) of typical ZSPMB topologies are derived. The equivalent reluctance force (ERF) is calculated and analyzed by combining the analytical method with the FEM. Finally, a Halbach ZSPMB is optimized based on the ELF model and the ERF suppression method. The validity of the proposed analysis method of ZSPMB including reluctance effect is verified by the FEM.

II. Variable Current Model and Equivalent Force of PMs

The field in the magnet is related by the constitutive relation [5]

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H} + \mu_0 \mathbf{M}_r \tag{1}$$

where μ_0 is the permeability of air, $\mathbf{B}$ is the magnetic flux density, $\mathbf{H}$ is the magnetic field strength, μ_r is the relative permeability of PM, and $\mathbf{M}_r$ is the residual magnetization of PM. Then, the field equation can be equivalently expressed as

$$\nabla \times \frac{\mathbf{B}}{\mu_0} = \nabla \times (\mu_r \mathbf{H} + \mathbf{M}_r). \tag{2}$$

As shown in Fig. 1(a), $\mathbf{n}$ and $\mathbf{t}$ are the normal unit vector and the tangential unit vector on the surface of PM, respectively, and the fields on both sides are written as $\mathbf{B}_1$ and $\mathbf{B}_2$. Assume that $\mathbf{M}_r$ is uniform, (2) exists non-zero value only on the surface because there is no conduction current in the magnet. Applying the boundary condition of the tangential component of field strength, i.e., $\mathbf{H}_{1t} = \mathbf{H}_{2t}$, (2) can be written as

$$\begin{aligned} \nabla \times \frac{\mathbf{B}}{\mu_0} &= (\mu_r \mathbf{H}_2 - \mathbf{H}_1) \times \mathbf{n} + \mathbf{M}_r \times \mathbf{n} \\ &= (\mu_r - 1)\mathbf{H}_{1t} \times \mathbf{n} + \mathbf{M}_r \times \mathbf{n} \\ &= \mathbf{J}_\mu + \mathbf{J}_r \end{aligned} \tag{3}$$

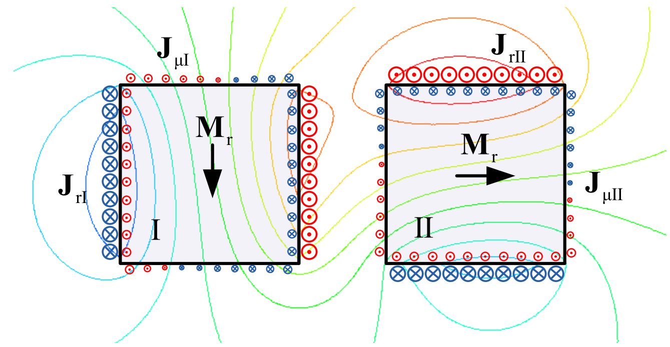

Fig. 2. Field and variable current distributions of PM bearing.

where

$$\mathbf{J}_r = \mathbf{M}_r \times \mathbf{n} \tag{4}$$

and

$$\mathbf{J}_\mu = (\mu_r - 1)\mathbf{H}_{1t} \times \mathbf{n} \tag{5}$$

are called the equivalent magnetization current (EMC) and the equivalent reluctance current (ERC), respectively. Thus, as shown in Fig. 1(b), the PM can be replaced by the surface currents $\mathbf{J}_r$ and $\mathbf{J}_\mu$ in air.

To illustrate (3), the 2-D field and equivalent current distributions of PM bearing are shown in Fig. 2, which contains two magnets with a rectangular cross section and perpendicular magnetization. The magnets are represented by the equivalent currents distributing on the surfaces with units A/m. The $\mathbf{J}_{\mathrm{rI}}$ and $\mathbf{J}_{\mathrm{rII}}$ are the EMCs of magnets I and II, respectively. According to (4), the EMC of the magnet consists of two constant current sheets whose normal vectors are parallel to the magnetization direction. $\mathbf{J}_{\mu\mathrm{I}}$ and $\mathbf{J}_{\mu\mathrm{II}}$ are the ERCs of magnets I and II, respectively. According to (5), the ERC of the magnet is composed of four current sheets, which are variable with the tangential component of field strength on the boundaries in air. Because μ_r is close to 1, the ERC is usually ignored in the analysis of high stiffness PM bearing. In [4], the equivalent current model is built based on the average operating point of PMs to consider the effect of μ_r in the analysis of ZSPMB; however, the complex distribution of the ERC is ignored. In this paper, the ZSPMB is analyzed based on the variable current model according to (3), in which the magnetic permeability is effectively considered and its influences on the magnetic field and force are attributed to the reluctance effect.

According to the variable current model, the force exerted between PMs can be expressed as the superposition of the forces produced by equivalent currents and is given by

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_{\mathbf{J}_{\mathrm{rI}}\mathbf{J}_{\mathrm{rII}}} + \mathbf{F}_{\mathbf{J}_{\mathrm{rI}}\mathbf{J}_{\mu\mathrm{II}}} + \mathbf{F}_{\mathbf{J}_{\mu\mathrm{I}}\mathbf{J}_{\mathrm{rII}}} + \mathbf{F}_{\mathbf{J}_{\mu\mathrm{I}}\mathbf{J}_{\mu\mathrm{II}}} \\ &= \mathbf{F}_R + \mathbf{F}_L \end{aligned} \tag{6}$$

where $\mathbf{F}_{\mathbf{J}_{\mathrm{rI}}\mathbf{J}_{\mathrm{rII}}}$ is the force between $\mathbf{J}_{\mathrm{rI}}$ and $\mathbf{J}_{\mathrm{rII}}$, $\mathbf{F}_{\mathbf{J}_{\mathrm{rI}}\mathbf{J}_{\mu\mathrm{II}}}$ is the force between $\mathbf{J}_{\mathrm{rI}}$ and $\mathbf{J}_{\mu\mathrm{II}}$, $\mathbf{F}_{\mathbf{J}_{\mu\mathrm{I}}\mathbf{J}_{\mathrm{rII}}}$ is the force between$\mathbf{J}_{\mu\mathrm{I}}$ and $\mathbf{J}_{\mathrm{rII}}$, and $\mathbf{F}_{\mathbf{J}_{\mu\mathrm{I}}\mathbf{J}_{\mu\mathrm{II}}}$ is the force between $\mathbf{J}_{\mu\mathrm{I}}$ and $\mathbf{J}_{\mu\mathrm{II}}$

$$\mathbf{F}_L = \mathbf{F}_{\mathbf{J}_{\mathrm{rI}}\mathbf{J}_{\mathrm{rII}}} \tag{7}$$

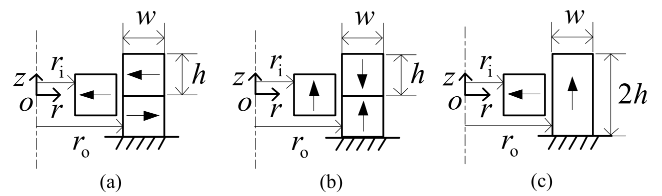

Fig. 3. Typical ZSPMB topologies. (a) Radial magnetization (R-ZSPMB). (b) Axial magnetization (A-ZSPMB). (c) Perpendicular magnetization (P-ZSPMB).

and

$$\mathbf{F}_R = \mathbf{F}_{\mathbf{J}_{\mathrm{rI}}\mathbf{J}_{\mu\mathrm{II}}} + \mathbf{F}_{\mathbf{J}_{\mu I}\mathbf{J}_{\mathrm{rII}}} + \mathbf{F}_{\mathbf{J}_{\mu\mathrm{I}}\mathbf{J}_{\mu\mathrm{II}}}. \tag{8}$$

Both $\mathbf{F}_L$ and $\mathbf{F}_R$ can be regarded as Lorenz forces. However, due to the complex distributions of ERC, $\mathbf{F}_R$ is difficult to compute using Lorenz law. For $\mathbf{F}_R$ results from the reluctance effect, it is considered as reluctance force and calculated with a hybrid analytical-FEM method in this paper. Hence, $\mathbf{F}_L$ and $\mathbf{F}_R$ are called the ELF and the ERF, respectively. Compared with the conventional reluctance force, which is an attractive force exerted by magnetic field on ferromagnetic material, $\mathbf{F}_R$ has different characteristics as described below.

III. Calculation of Magnetic Field and Force

The parameter analysis and optimization of ZSPMB involve a large number of high accuracy magnetic field and force calculations. In order to improve the accuracy and efficiency of the calculation, this paper proposes a hybrid analytical-FEM approach for analyzing the ZSPMB, in which the ELF is analytically calculated based on the Biot–Savart law, elliptic integral, and Lorenz law, and then the ERF is obtained by combining the FEM.

Three typical ZSPMB topologies using rings with radial, axial, and perpendicular magnetizations, respectively, are shown as Fig. 3. The cylindrical coordinates are used with the rings, which are centered with respect to the z-axis. To calculate the ELF, the ring magnets are replaced by the EMC flowing through the circular coils located at their faces. For an axially magnetized ring magnet, it can be represented by the cylindrical surface currents which correspond to its inner and outer faces. For a radially magnetized ring magnet, it can be represented by the ring surface currents which correspond to its upper and lower faces. Applying the Biot–Savart law, the magnetic field $\mathbf{B}$ at any position in the rz plane generated by the surface current $\mathbf{J}_r$ at the position (r_1, θ, z_1) can be calculated with the following equation:

$$\begin{aligned} d\mathbf{B}(r,z) &= \frac{\mu_0 \mathbf{J}_r \times \rho}{4\pi|\rho|^3} dS \\ &= \frac{\mu_0 J_r((z-z_1)(\cos\theta\mathbf{r}-\sin\theta\theta)+(r\cos\theta-r_1)\mathbf{z})}{4\pi\left(r^2+r_1^2-2rr_1\cos\theta+(z-z_1)^2\right)^{\frac{3}{2}}} dS \end{aligned} \tag{9}$$

where ρ is the displacement vector from the position of $\mathbf{J}_r$ to the position where the field $\mathbf{B}$ is being calculated and S is the area of surface current.

Due to the symmetry, only the z-component of the magnetic force of the ZSPMB is analyzed. Namely, it is only necessary to compute the r-component of $\mathbf{B}$, which can be expressed as follows for a cylindrical surface current:

$$\begin{aligned} &B_{r,\mathrm{axial}}(r,z) \\ &= \int_0^{2\pi}\int_{z_{1-}}^{z_{1+}} \frac{\mu_0 J_r r_1 (z-z_1)\cos\theta_1}{4\pi(r^2+r_1^2-2rr_1\cos\theta_1+(z-z_1)^2)^{\frac{3}{2}}} dz_1 d\theta_1 \\ &= \int_0^{\pi} \frac{\mu_0 J_r r_1 \cos\theta_1}{2\pi\sqrt{r^2+r_1^2-2rr_1\cos\theta_1+(z-z_1)^2}} d\theta_1\Big|_{z_{1-}}^{z_{1+}} \\ &= \int_0^{\pi/2} \frac{-\mu_0 J_r r_1 \cos 2\phi}{\pi\sqrt{r^2+r_1^2+2rr_1\cos 2\phi+(z-z_1)^2}} d\phi\Big|_{z_{1-}}^{z_{1+}} \\ &= \frac{-\mu_0 J_r r_1}{\pi}\int_0^{\pi/2} \frac{1-2\sin^2\phi}{\sqrt{k_1^2-4rr_1\sin^2\phi}} d\phi\Big|_{k_{1-}}^{k_{1+}} \\ &= \frac{-\mu_0 J_r r_1}{\pi}\left(\frac{k_1\mathbf{E}\left(\frac{4rr_1}{k_1^2}\right)}{2rr_1} + \left(k_1^{-1}-\frac{k_1}{2rr_1}\right)\mathbf{K}\left(\frac{4rr_1}{k_1^2}\right)\right)\Bigg|_{k_{1-}}^{k_{1+}} \end{aligned} \tag{10}$$

where

$$k_1 = \sqrt{r^2+r_1^2-2rr_1+(z-z_1)^2} \tag{11}$$

$$\theta_1 = \pi - 2\phi \tag{12}$$

$$\mathbf{K}(m) = \int_0^{\pi/2} \frac{1}{\sqrt{1-m\sin^2\phi}} d\phi \tag{13}$$

and

$$\mathbf{E}(m) = \int_0^{\pi/2} \sqrt{1-m\sin^2\phi}\, d\phi \tag{14}$$

are the first and second complete elliptic integrals, respectively. The numerical integration is avoided by using elliptic integral and, therefore, it is computationally fast.

The r-component of $\mathbf{B}$ produced by a ring surface current can be expressed as follows:

$$\begin{aligned} &B_{r,\mathrm{radial}}(r,z) \\ &= \int_{r_{1-}}^{r_{1+}}\int_0^{2\pi} \frac{\mu_0 J_r r_1 (z-z_1)\cos\theta_1}{4\pi\left(r^2+r_1^2-2rr_1\cos\theta_1+(z-z_1)^2\right)^{\frac{3}{2}}} d\theta_1 dr_1 \\ &= \frac{-\mu_0 J_r r_1 (z-z_1)}{\pi}\int_{r_{1-}}^{r_{1+}}\int_0^{\pi/2} \frac{1-2\sin^2\phi}{\left(k_1^2-4rr_1\sin^2\phi\right)^{\frac{3}{2}}} d\phi dr_1 \\ &= k_2\int_{r_{1-}}^{r_{1+}}\int_0^{\pi/2}\left(\frac{k_1^2/2rr_1}{\left(1-\frac{4rr_1}{k_1^2}\sin^2\phi\right)^{\frac{1}{2}}} + \frac{1-k_1^2/2rr_1}{\left(1-\frac{4rr_1}{k_1^2}\sin^2\phi\right)^{\frac{3}{2}}}\right) d\phi dr_1 \end{aligned}$$

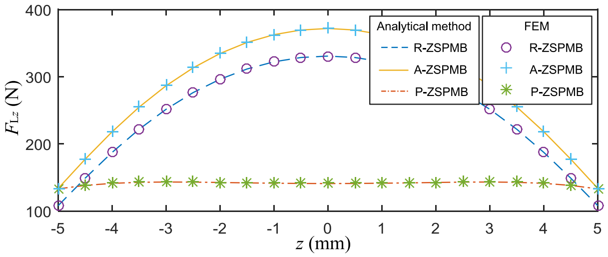

Fig. 4. Comparison of force-displacement curves for ELFs calculated with the analytic method and FEM (with $\mu_r = 1$) for three typical ZSPMB topologies.

$$= k_2 \int_{r_{1-}}^{r_{1+}} \left(\frac{k_1^2 \mathbf{K}\left(\frac{4rr_1}{k_1^2}\right)}{2rr_1} + \left(1 - \frac{k_1^2}{2rr_1}\right) \Pi\left(\frac{4rr_1}{k_1^2}, \frac{4rr_1}{k_1^2}\right)\right) dr_1$$

$$= k_2 \int_{r_{1-}}^{r_{1+}} \left(\frac{k_1^2 \mathbf{K}\left(\frac{4rr_1}{k_1^2}\right)}{2rr_1} + \left(1 - \frac{k_1^2}{2rr_1}\right) \frac{\mathbf{E}\left(\frac{4rr_1}{k_1^2}\right)}{1 - 4rr_1/k_1^2}\right) dr_1 \tag{15}$$

where

$$k_2 = \frac{-\mu_0 J_r r_1 (z - z_1)}{\pi k_1^3} \tag{16}$$

$$\Pi(n, m) = \int_0^{\pi/2} \frac{1}{(1 - n\sin^2\phi)\sqrt{1 - m\sin^2\phi}} d\phi \tag{17}$$

is the third complete elliptic integral. It can be observed that the expression only owns one numerical integration.

Based on (10) and (15), the r-component of magnetic field B_r created by the outer rings can be obtained using superposition. Then, the z-component of the ELF, i.e., F_{Lz}, can be calculated by integrating B_r on the EMC of the inner one. The F_{Lz} can be written as follows:

$$F_{Lz,\text{radial}} = 2\pi J_r \int_{r_-}^{r_+} r(B_r(r, z_+) - B_r(r, z_-))dr \tag{18}$$

and

$$F_{Lz,\text{axial}} = 2\pi r J_r \int_{z_-}^{z_+} (B_r(r_+, z) - B_r(r_-, z))dz \tag{19}$$

for the inner ring with the radial magnetization and the axial magnetization, respectively.

Using (18) or (19), the ELF can be calculated for three typical ZSPMB topologies with MATLAB program, in which the complete elliptic integrals of the first and the second kind are computed with the MATLAB function *ellipke*. As an example, the inner radius r_i, width w, and height h are all 10 mm, and the outer radius r_o is 22 mm. The magnet material is ideal NdFeB35 with $M_r = 978\,802$ A/m and $\mu_r = 1$. For comparison, the finite element software ANSYS Maxwell is used to establish the 2-D axisymmetric FEMs consistent with Fig. 3. The percent error of finite element analysis is specified as 5×10^{-6} to obtain the smooth force-displacement curve. Using the parametric analysis function of ANSYS Maxwell, the forces at various locations along the z-axis in the range of interest are determined. As shown in Fig. 4, the analytical results are in good agreement with the FEM. It can be noticed that the zero stiffness points appear at $z = 0$ mm, where the maximum value of ELF is reached. Moreover, compared with the R-ZSPMB and A-ZSPMB, the working range of P-ZSPMB is significantly increased with more than half of the ELF loss. According to Lorentz law, a large external magnetic field gradient results in high stiffness when the current is constant. The magnetic field near the PM magnetic pole is strong; however, the magnetic field gradient is also large. It means that the force and stiffness of PM bearing have inherent strong coupling characteristics. The results show that the P-ZSPMB is more suitable for high-performance vibration isolation.

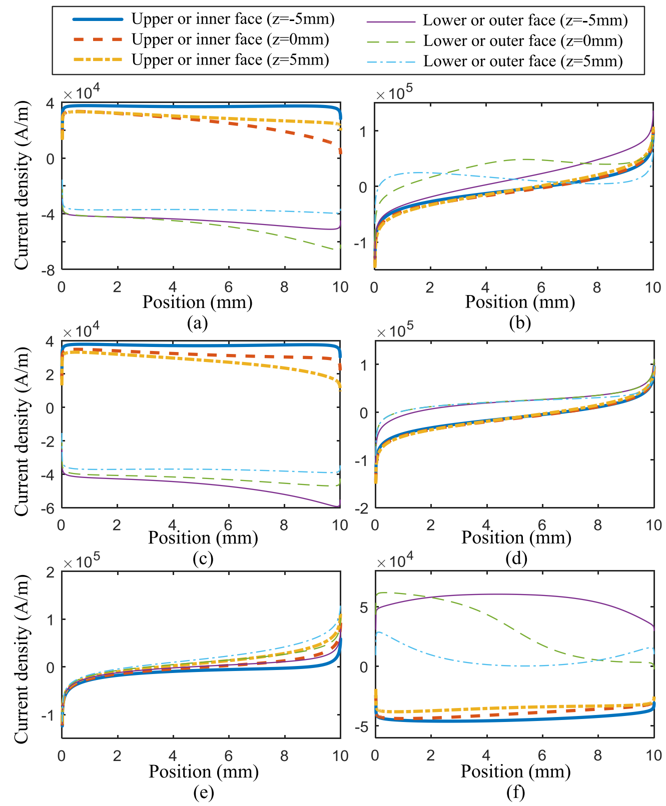

Fig. 5. Distributions of ERCs on boundaries. (a) Upper and lower faces of R-ZSPMB. (b) Inner and outer faces of R-ZSPMB. (c) Upper and lower faces of A-ZSPMB. (d) Inner and outer faces of A-ZSPMB. (e) Upper and lower faces of P-ZSPMB. (f) Inner and outer faces of P-ZSPMB.

In order to analyze the reluctance effect, as shown in Fig. 5, the ERCs are calculated using (5) based on the results of the finite element analysis with $\mu_r = 1.0998$. The abscissas represent the distance from the upper or inner end of the boundaries, and z is the vertical displacement of the inner ring. It can be found that the ERCs are unevenly distributed over all faces and approximately one order of magnitude less than the EMCs, i.e., equal to $\pm M_r$. Furthermore, the asymmetric distributions between the upper and lower faces vary complexly with the vertical displacement of the inner ring and the same for the inner and outer faces. To avoid the analytical calculation of the ERC, the related ERF can be obtained with a hybrid analytical-

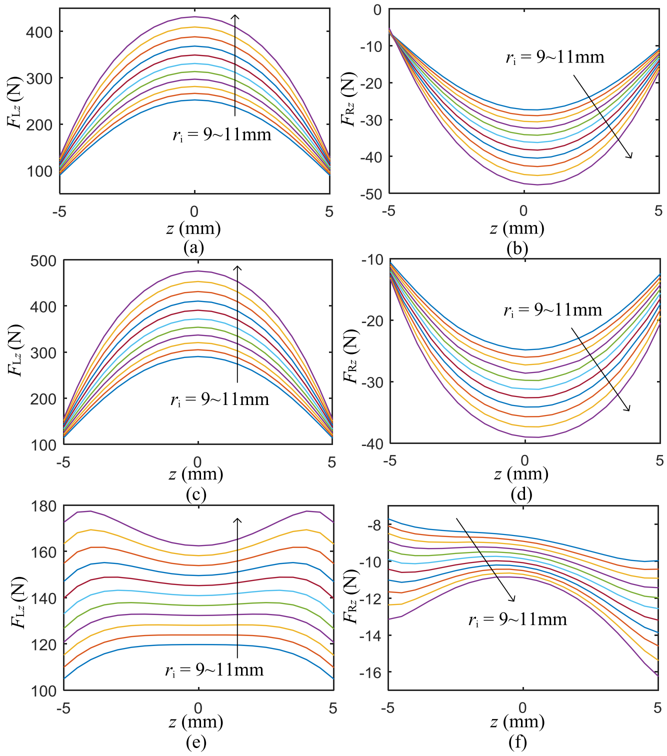

Fig. 6. Force-displacement curves for F_{Lz} and F_{Rz}. (a) F_{Lz} of R-ZSPMB. (b) F_{Rz} of R-ZSPMB. (c) F_{Lz} of A-ZSPMB. (d) F_{Rz} of A-ZSPMB. (e) F_{Lz} of P-ZSPMB. (f) F_{Rz} of P-ZSPMB.

FEM approach as follows:

$$F_{Rz} = F_z - F_{Lz} \tag{20}$$

where F_z and F_{Lz} are calculated with the FEM and the analytical method, respectively. Importing FEM results into MATLAB and calculating (20), the stiffness can be derived from the fitting curve of forces by using the polynomial curve fitting function *polyfit* and the polynomial derivative function *polyder* of MATLAB. Then, the vertical stiffness is expressed as follows:

$$K_z = K_{Rz} + K_{Lz} \tag{21}$$

where K_{Rz} and K_{Lz} are stiffness corresponding to the ERF and ELF, respectively.

The force-displacement curves for F_{Lz} and F_{Rz} with various r_i are shown in Fig. 6. It can be seen that, compared with F_{Lz}, F_{Rz} is about one order of magnitude smaller and in the opposite direction, and its force-displacement curve is deflected. The ERF is more complex than the conventional reluctance force, which is an attractive force exerted by magnetic field on ferromagnetic material. Due to the influence of F_{Rz}, the extreme point of F_z drifts from $z = 0$ mm, and the variation of F_z in the working range increases.

IV. Application in Optimization of Halbach Permanent Magnet Bearing

To illustrate the effectiveness of the method presented above, it is applied in the optimization of the cylindrical Halbach ZSPMB proposed in [4], whose cross-sectional configuration is shown in Fig. 7. The Halbach ZSPMB can be treated

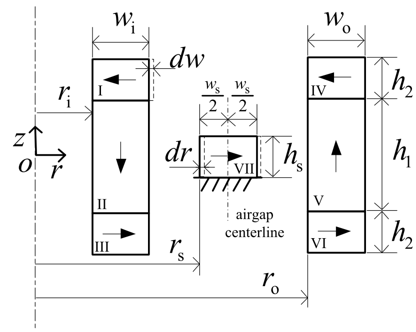

Fig. 7. Cross-sectional configuration. (Parameters dw and dr proposed by this paper are used to suppress ERF.)

TABLE I
Optimization Parameters

Parameters	Constraints	Initial value [4]	Optimal value	Unit
Inner radius of inner magnet ring (r_i)	[4,6]	4	4.2	mm
Width of inner magnet ring (w_i)	[4,8]	5	4.4	mm
Inner radius of outer magnet ring (r_o)	[28,33]	31	31.2	mm
Width of outer magnet ring (w_o)	[1,5]	5	2.2	mm
Radius of stator (r_s)	[15,20]	17	16.4	mm
Thickness of stator (h_s)	[4,10]	4	5.8	mm

as a combination of the R-ZSPMB and the P-ZSPMB. Its main design objectives are to generate vertical force $F_z = 40$ N with vertical stiffness $K_z < 100$ N/m in ± 1 mm vertical working range. The used magnet material is NdFeB35. The selected optimization parameters are given in Table I, and the fixed thicknesses of Halbach arrays are shown in Table II. The stator, i.e., the ring VII, is placed in the middle of the airgap between the inner and outer rings, thus its width can be expressed as $w_s = r_o + r_i + w_i - 2r_s$. In addition, the parameters *dw* and *dr*, which indicate the width variation of the ring I and the inner radius variation of the stator, respectively, are proposed by this paper for the ERF suppression. The optimization can be summarized in the following three steps: First, the basic optimization is carried out based on the analytical model of F_{Lz}. Second, F_{Rz} is calculated and analyzed with the hybrid analytical-FEM approach. Third, the fixed step search for *dw* and *dr* is fulfilled to suppress the ERF.

The F_{Lz} is computed analytically using (18). Then, the stiffness K_{Lz} is calculated from the fitting curve of F_{Lz} by linear derivation. The maximum absolute value of K_{Lz} in the working range is used as the object of minimization, and the objective function is expressed as follows:

$$F(r_i, r_o, r_s, w_i, w_o, h_s) = \min(\max(|K_{Lz}|)). \tag{22}$$

The optimization is carried out with a genetic algorithm of MATLAB Optimization Tool. The optimized parameters are shown in Table I.

Fig. 8 shows the comparisons between the initial design and the optimal design. As shown in Fig. 8(b), it can be seen that F_{Lz} versus displacement curve of the optimal design has three extreme points in the working range. One extreme point

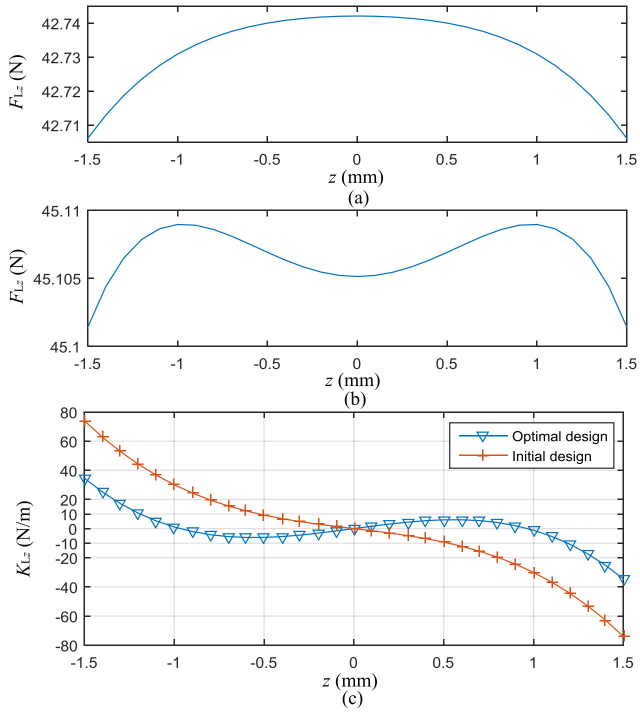

Fig. 8. Comparisons between initial design and optimal design about F_{Lz} and K_{Lz}. (a) F_{Lz} of the initial design. (b) F_{Lz} of optimal design. (c) K_{Lz} of the initial design and optimal design.

TABLE II

FIXED PARAMETERS

Parameters	Value [4]	Unit
Thickness of middle magnet rings (h_1)	20	mm
Thickness of top and bottom magnet rings (h_2)	4	mm

is at $z = 0$ mm, and the other two extreme points are near $z = -1$ mm and $z = 1$ mm, respectively. Accordingly, K_{Lz} versus displacement curve of the optimal design has three zero stiffness points. As shown in Fig. 8(c), K_{Lz} of the optimal design decreases significantly compared with the initial design, which has only one extreme point as shown in Fig. 8(a). In other words, the optimal design can enlarge the working range with the same stiffness as the initial design.

Further, the optimal design is analyzed by the FEM to establish the ERF model. The 2-D axisymmetric FEM is consistent with Fig. 7. Because the stiffness of ZSPMB is very sensitive to the force calculation error, the influences of the number of FEM mesh element on the calculation errors of force and stiffness are analyzed. As shown in Fig. 9, the force calculation error on the order of mN will greatly affect the stiffness calculation accuracy since the stiffness is on the order of mN/mm. The smooth force-displacement curve can be obtained by using fine mesh. Using the FEM calculation result of 2 070 000 elements as an accurate value, the stiffness calculation error can be reduced from 243% to 31% by increasing the number of elements from 151 000 to 732 000. However, it takes about 8 h to calculate a force-displacement curve in 2 mm stroke with 0.1 mm step size and 732 000 elements. Therefore, the FEM is not suitable for

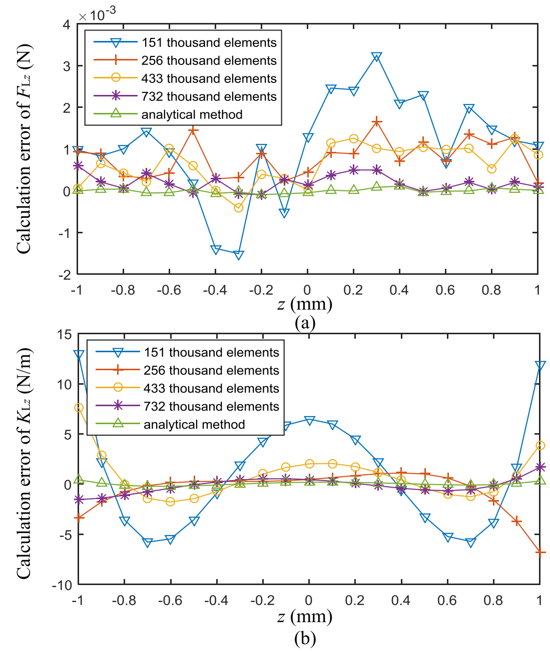

Fig. 9. Influences of FEM mesh element number on force and stiffness calculation errors. (a) F_{Lz}. (b) K_{Lz}.

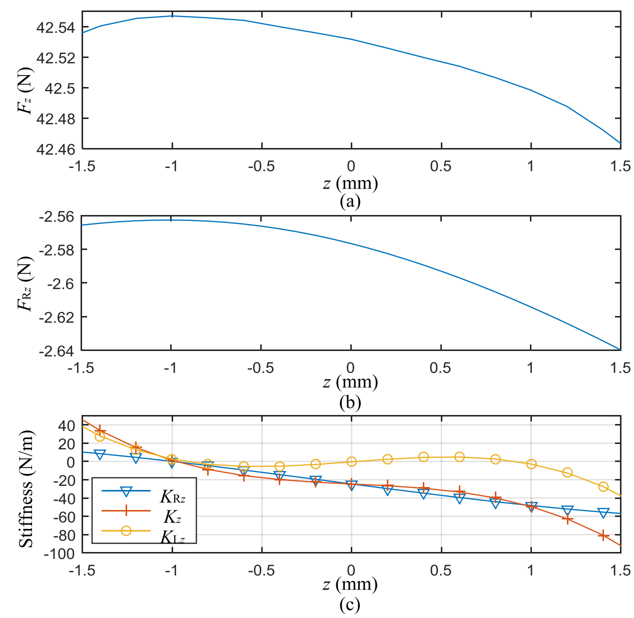

Fig. 10. Force and stiffness of optimal design considering the reluctance effect. (a) F_z. (b) F_{Rz}. (c) Stiffness K_{Rz}, K_{Lz}, and K_z.

the parametric studies and optimization of ZSPMB. In the hybrid analytical-FEM approach, the analytical analysis and optimization are mainly carried out by the MATLAB program, and the calculation of one force-displacement curve only takes about 3.5 s and the stiffness calculation error is about 8%. Moreover, only one force-displacement curve is calculated by the FEM throughout the optimization process. The time-consuming of the hybrid analytical-FEM approach decreases significantly and its effectiveness is demonstrated below.

The influences of the reluctance effect on the optimal design are shown in Fig. 10, where F_z is obtained by the FEM,

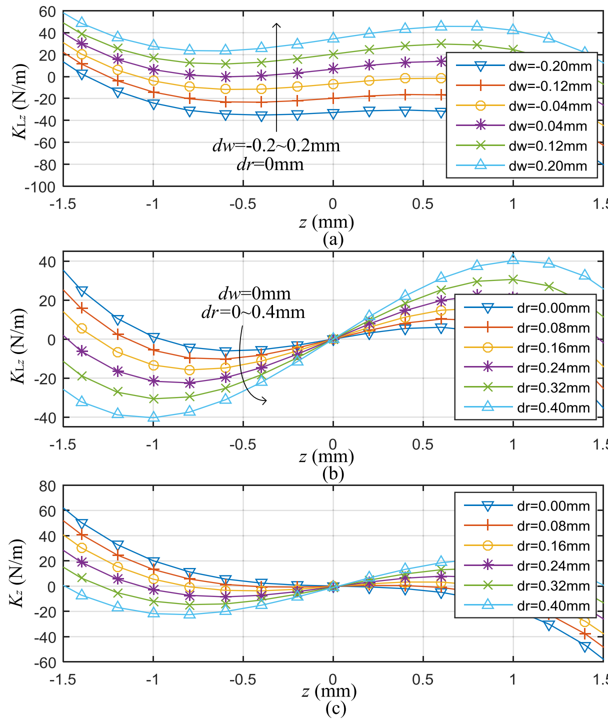

Fig. 11. Analysis of ERF suppression effects of dr and dw with analytical model. (a) Drift effect of dw on K_{Lz} ($dr = 0$ mm, $dw = -0.2$–0.2 mm). (b) Deflection effect of dr on K_{Lz} ($dw = 0$ mm, $dr = 0$–0.4 mm). (c) ERF suppression effect on K_z ($dw = 0.14$ mm, $dr = 0$–0.4 mm).

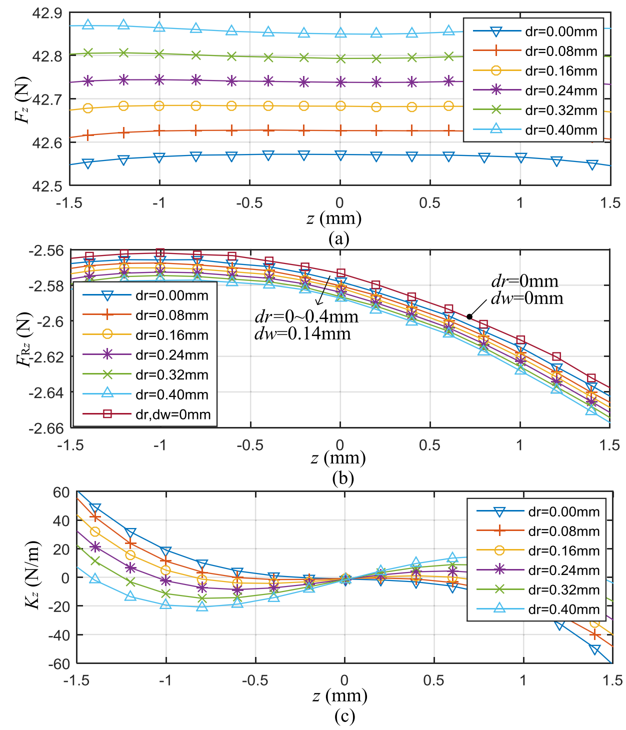

Fig. 12. ERF suppression effect calculated with FEM. (a) Effect of dr on F_z ($dw = 0.14$ mm, $dr = 0$–0.4 mm). (b) Effect of dr on F_{Rz} ($dw = 0.14$ mm, $dr = 0$–0.4 mm). (c) ERF suppression effect on K_z ($dw = 0.14$ mm, $dr = 0$–0.4 mm).

F_{Rz} is calculated by (20), and the stiffness is computed by linear derivation of the force. It can be observed that the reluctance effect has little influence on the bearing capacity but has a significant influence on the stiffness versus displacement curve. As shown in Fig. 10(c), K_{Rz} causes K_z versus displacement curve to deflect and drift. Consequently, compared with K_{Lz}, the zero stiffness point of K_z drifts nearly -1 mm and the stiffness variation of K_z in the working range increases obviously. Therefore, it is necessary to effectively suppress K_{Rz}.

The parameters *dw* and *dr* indicate the width variation of the ring I and the inner radius variation of the stator, respectively. The drift effect of *dw* and the deflection effect of *dr* on K_{Lz} versus displacement curve are shown in Fig. 11(a) and (b), respectively. It can be seen that K_{Lz} is sensitivity to the variations of *dr* and *dw*. Therefore, an optimal K_{Lz} can be obtained by varying *dr* and *dw* in a small range with fixed step to cancel K_{Rz}, which can be regarded as constant because the dimensional variations have a much greater impact on ELF than ERF. It means that K_z can be calculated from

$$K_z = K_{Lz} + K_{Rz0} \tag{23}$$

where K_{Lz} is computed analytically and K_{Rz0} is obtained by combining the analytical method with the FEM at $dr = 0$ mm, $dw = 0$ mm. The maximum absolute value of K_z in the working range is used as the object of minimization. The optimization can be carried out with a fixed step search. The suppression effect of the ERF on K_z at $dw = 0.14$ mm, $dr = 0$–0.4 mm is shown in Fig. 11(c). Comparing K_{Lz} in Fig. 8(c) with K_z in Fig. 11(c), one can find that the ERF can be suppressed effectively.

TABLE III
OPTIMAL DESIGN RESULTS FOR FOUR BEARING CAPACITIES

Parameters	F_z=42.7N	F_z=52.0N	F_z=59.2N	F_z=72.7N	Unit
r_i	4.2	4.8	4.6	5	mm
w_i	4.4	5.1	7.4	5.6	mm
r_o	31.2	31	30.6	30.8	mm
w_o	2.2	3.4	2.1	1.5	mm
r_s	16.4	17.7	19.3	17.3	mm
h_s	5.8	6.2	8.4	8.6	mm
dw	0.15	0.24	0.23	0.13	mm
dr	0.2	0.15	0	0.32	mm

As shown in Fig. 12, the ERF suppression effects can be further verified with the force and stiffness calculated by the FEM. The effects of *dr* on F_z and F_{Rz} at $dw = 0.14$ mm are shown in Fig. 12(a) and (b), respectively. One can find from Fig. 12(b) that small dimensional variations cause a slight drift of F_{Rz} versus displacement curves but the corresponding stiffness remain the same. It can be seen that K_z calculated with the FEM shown in Fig. 12(c) are in good agreement with the analytical calculations shown in Fig. 11(c).

To further illustrate the effectiveness of the proposed method, the optimal design results for four bearing capacities, including 42.7N, 52.0N, 59.2N, and 72.7N, are shown in Table III. The corresponding K_z versus displacement curves are shown in Fig. 13. The results show that the maximum absolute value of K_z in the working range increases with the bearing capacity but the resonance frequencies remain at the

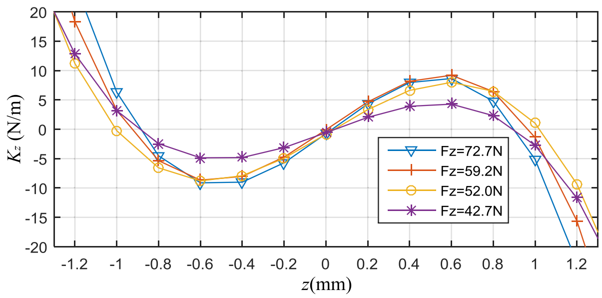

Fig. 13. K_z versus displacement curves calculated with FEM for four optimal designs.

sub-hertz level according to the equation

$$f = \sqrt{K/M}/2\pi \tag{24}$$

where K is stiffness and M is the mass that can be obtained by dividing force by the gravitational constant.

V. Conclusion

In this paper, a hybrid analytical-FEM approach is proposed to analyze and optimize the ZSPMB including the reluctance effect. The force exerted between PMs is modeled as the superposition of the ELF and the ERF. The ELF can be solved quickly in an analytical form. The ERF is obtained by combining the FEM, in which the effect of magnetic permeability is fully considered. Three typical ZSPMB topologies are investigated. The results show that the ERF is much smaller than the ELF but it leads to the deflection of the force-displacement curve and the significant increase in stiffness. A Halbach ZSPMB is optimized with the proposed approach, in which only one force-displacement curve is calculated by the FEM throughout the optimization process and the time-consuming decreases significantly. The stiffness-displacement curve corresponding to the ELF is shifted and deflected by adjusting the inner magnet ring width and the stator radius, respectively, to suppress the ERF. The results show that the stiffness variation in the working range is reduced obviously.

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